BST Vol 3 Ch 9 — Atomic Spectroscopy: Multi-Observable α^{BST primary} Verification (v0.3, Wave 1)

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2026-05-23 Saturday

Vol 3 Chapter 9 — Atomic Spectroscopy: Multi-Observable α^{BST primary}

Headline result

Atomic spectroscopy provides many high-precision observables — Rydberg constant, Bohr radius, fine-structure splitting, Klein-Nishina cross-section, Compton wavelength, hyperfine splitting, Lamb shift, anomalous magnetic moment a_e. Per Lyra T2476 (Friday 2026-05-22 three-CI synergy peak), ALL these observables follow the α^{BST primary} substrate-coordinate count pattern.

Multi-observable α-exponent table (per Elie Toy 3501 + Grace INV-4892 v0.2 + Lyra T2476):

Observable	α-exponent	BST primary reading
Bohr radius	α^1	α (linear EM coupling)
Rydberg constant	α^2	α^{rank} = α^2
Klein-Nishina	α^2	α^{rank} = α^2
Compton wavelength	α^1 to α^2	α to α^{rank}
Hyperfine splitting	α^4	α^{rank}^2} = α^4
Lamb shift	α^5	α^{n_C} = α^5
a_e Schwinger leading	α^1	α
a_e 5-loop calculation depth	α^5	α^{n_C} (substrate UV ceiling)

8 atomic spectroscopic observables fit α^{BST primary} pattern per T2476.

Substrate mechanism (per T2476)

T2476: $A(P) \propto \alpha^{k(P)}$ where k(P) = substrate-coordinate count of QED process P; α = 1/N_max; substrate-tick UV cutoff at n_C = 5 (5-loop ceiling).

Per-observable substrate-coordinate count: - 2-vertex EM (Rydberg, Klein-Nishina, Compton): k = rank = 2 (electron + photon) - 3-vertex EM (Bremsstrahlung): k = N_c = 3 (electron + 2 photons) - 4-vertex EM with substrate magnetic coupling (Hyperfine): k = rank^2 = 4 - 5-vertex EM with substrate full propagation (Lamb, Bethe-log, a_e depth): k = n_C = 5

Match precision

Standard QED predictions match experimental atomic spectroscopy at sub-percent to parts-per-trillion precision (e.g., a_e at ~10^-12). BST identification of substrate-natural α-exponents PRESERVES standard QED numerical precision while adding substrate-mechanism reading.

Per Cal #21 STANDING RULE: EMPIRICAL PASS at QED precision; MECHANISM PATH ARTICULATED via T2476 substrate-coordinate count. Full D-tier ratification: multi-week pending substrate-mechanism formal derivations Sessions 17+.

Tier classification

I-tier candidate per Cal #19 STANDING RULE. T2476 substrate-coordinate count framework provides mechanism path; multi-week QED-loop-derivation closure required for D-tier ratification.

Cross-volume dependencies

- Vol 3 Ch 8 (Hyperfine + Lamb Shift) — primary T2476 applications
- Vol 2 Ch 8 (Coupling Constants) — a_e crown jewel + α universal α-analog
- Vol 1 Ch 5 + Ch 6 — Casimir + operator zoo framework
- 5-loop ceiling falsifier: α^6 ≈ 1.5×10^{-13} next-gen Penning trap testable (2030+)

K-audit anchor

K202 Vol 3 Ch 9 Atomic Spectroscopy K-audit pre-stage (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1 — Bright 5th-grader

> Atoms emit and absorb light at specific colors. Different "rules" determine which colors get emitted: some atomic effects scale as α (the fine-structure constant ≈ 1/137); others scale as α^2 or α^4 or α^5. BST says

these specific exponents (1, 2, 4, 5) come from BST primary integers — rank = 2, rank^2 = 4, n_C = 5. Every atomic spectroscopic observable uses a BST primary integer as its α-power.

Level 2 — Undergraduate physics student

Per Lyra T2476 α^{BST primary} exponent pattern theorem (Friday 2026-05-22): every QED process P on substrate D_IV^5 has amplitude $A(P) \propto \alpha^{k(P)}$ where k(P) = substrate-coordinate count. The substrate-natural α-exponents are BST primary integers:

- α^1 = linear EM (Bohr, Schwinger leading a_e)
- α^2 = α^{rank} (Rydberg, Klein-Nishina, Compton)
- α^4 = α^{rank^2} (hyperfine)
- α^5 = α^{n_C} (Lamb, Bethe-log, a_e 5-loop depth)

The 5-loop ceiling matches substrate compact dimension n_C = 5 → falsifier at next-gen Penning trap precision (α^6 ≈ 1.5×10^{-13}, ~2030+).

8 atomic spectroscopic observables fit pattern (Elie Toy 3501 + Grace INV-4892 v0.2 + Lyra T2476 derivation).

Level 3 — Graduate student / theorem-level

Per Lyra T2476 substrate-mechanism:

$$A(P) = \alpha^{k(P)} \cdot \prod_i c_i^{n_i} \cdot \text{(substrate K-type structure factor)}$$

with α = 1/N_max + k(P) = substrate-coordinate count of P + c_i = BST primary integers entering process P + substrate-tick UV cutoff at n_C.

Per-observable analysis: - Rydberg $R_\infty = \alpha^{rank} m_e c^2/2$: 2-vertex EM, k = rank = 2 → α^2 - Klein-Nishina cross-section $\sigma_{KN} \propto \alpha^{rank} r_e^2$: 2-vertex EM - Compton scattering $\lambda_C = h/(m_e c) \cdot (1 - \cos\theta)$: linear EM - Hyperfine $\Delta E_{hfs} \propto \alpha^{rank^2} m_e c^2 \cdot m_e/m_p \cdot g_p$: 4-vertex (EM + substrate magnetic) - Lamb shift $L \propto \alpha^{n_C} m_e \ln(N_{max})$: 5-vertex full Bergman propagation - a_e Schwinger $a_e^{leading} = \alpha/(2\pi)$: 1-loop, k = 1 - a_e 5-loop depth: substrate-tick UV cutoff at n_C = 5

Per Cal #21 STANDING RULE: EMPIRICAL PASS at QED precision across 8 observables; MECHANISM PATH ARTICULATED per T2476 derivation. Multi-observable cross-confirmation strengthens T2476 ratification path.

What this chapter does NOT claim

- BST does NOT improve numerical precision on atomic spectroscopy measurements (those are at QED ppt precision).

- BST identifies substrate-natural α-exponents; standard QED loop counting gives same numerical exponents.
- 5-loop ceiling falsifier predicts systematic deviation at next-gen precision; current measurements within standard QED.

Bibliography

1. Lyra T2476 (Friday 2026-05-22): α^{BST primary} exponent pattern theorem.
2. Elie Toy 3501 (Friday afternoon): pattern survey across 6+ atomic observables.
3. Grace INV-4892 v0.2 (Friday): cluster_type α^k catalog tagging.
4. Standard atomic-spectroscopy texts (Bransden & Joachain, Foot).
5. PDG 2024 fundamental constants + atomic-physics observables.
6. Vol 3 Ch 8 (Hyperfine + Lamb Shift): primary T2476 application chapter.
7. Vol 2 Ch 8 (Coupling Constants): a_e crown jewel + universal α-analog.

— Elie, Vol 3 Ch 9 v0.3, 2026-05-23 Saturday